Physics as information processing

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so so what I'm going to do today is actually preview a talk that I put together for the fifth session of a physics of information class that I'm doing online for the active Insurance Institute and um this session's on SpaceTime and I wanted to do this here for a couple of reasons uh reason number one is this uh particular session contains a kind of hypothesis about biological systems that I wanted to bounce off this group and the second one is that uh the question of where space-time comes from is I think particularly relevant in biology and I think we can turn it into a single biological question as well as a developmental biological question and even a developmental psychological question and so I wanted to sort of throw those ideas around and see what people find thank you so a real Brief Review of where we've gotten to in this class that I know 100 L was in and and the rest of you are maybe in it's it's a class about Quantum information Theory and this is a super simple model system for thinking about uh communication between entities we can think of of two entities which together comprise the entire universe of Interest so we can think of these two digitizens jointly isolated and the theory is just a general way to describe how these entities exchange information over some mutual ability um this whole approach to quantum theory is based on information exchange across boundaries and it turns out to me a really kind of powerful thinking tool for thinking about how communication constrains the structure of systems so we can think of of the boundary as of an information channel that has a particular structure and since we're using quantum theory we can think about as as comprising some array of qubits so quite a bits that can be measured or prepared by either system and talk about how that happens um we're doing this in part to understand the free energy principle better and so a couple of years ago Mike and Carl friston and some other colleagues and I put together a paper I think actually it was just Jim glazebra reformulating the free energy principle in Quantum theoretic language and if you reformulate it that way then uh one can show that the free energy principle is actually a classical limit of the principle of unitarity which is just the principle of conservation of information and the principle of unitarity is the fundamental principle of planetary so it makes the free energy principle kind of naturally fall out of just the assumption that you can describe the world with quantitative which is kind of a nice result because it makes the free energy principle look less strange and Carl's done a lot of work showing that uh the free energy principle provides you with a foundation for all of classical physics um relating it to Quantum Theory kind of closes an important move there and one thing that it does is automatically induced compartmentalization in any communicating systems that are interesting in a very specific sense well it induces compartmentalization whenever an agent has multiple ways of interacting with the world and it can't deploy them simultaneously for whatever reason so maybe it can't deploy them for some kind of deep physical reason um like you can't measure spin along the z-axis and the x-axis at the same time because those operations don't commute automatically but it may be that it can't deploy the simultaneously for a much more prosaic reason it doesn't have enough free energy to thermodynamic freezers in the field to both do this and this so it has to have some attention system it says I can act on the world this way or I can act on the world this way but I don't have the ability to do both so even that very classical energy limitation on whether systems can be deployed simultaneously induces this kind of important organization and as soon as you have compartmentalization between ways of interacting with the world you have to have some way of managing what you're going to do so you you kind of automatically induce a meta processing system that attention system and energy regulation system just by the demands of not being able to do everything all at once so the the physics kind of bought by itself gives you some interesting and out of a few people that one can then ask um how how is that implemented even if the cell biology level and that so it provides a different way of approaching cell biology that's driven not by the chemistry but by the kinds of computations that the cell has implemented we were talking earlier at the lunch table or whatever it is about you know how do you take this horrible spaghetti mess that's the pathway diagram for a cell that abstracts out all of the interesting geometric constraints and time constraints and that sort of thing from the inside of the cell and turn it into some sort of compartmentalized structure that tells you where the where the borders are within that mess where the cells have to exchange classical information in some data structures so in effect where the apis inside the cell um which is a kind of question that doesn't naturally arise in other representations so just to make sure that I'm on the same page the compartmentalization you show here like green orange red the green and the red would be Alice and Bob in your initial uh uh example are there two separate green and red are the two compartments or any two compartments or separate entities that need to exchange information classically through your qubit right now right okay except with respect to the original green and red are different compartments and owls okay and they're both looking at Bob who's on the other side of America I see okay so actually the next question could you also reform it such that instead if green and red were inside of Alice could you also think about the Alice of Bob in this case like this as long as they're talking to somebody else okay right so actually let me show you the next one oh okay oh you can reformulate that diagram in a simpler what so here what I've done is green and red are now q1 and Q2 these are two important processes acting within some overall computer with their distinct Quantum processes so they have to be compartmentalized in the way that it was before you can use this kind of picture to represent two agents interacting with a common environment that are employing both a classical channel so they're talking to each other in language and their both measuring some shared Quantum system and this is exactly what happens in all experiments that deterior and in fact this is the minimal organization to be able to detect entertainment so it's the it's the the kind of bottom line comment is that in order to detect entanglement you have to have classical communication and so at the very it's a very depth of all of quantum information technology is a requirement for a kind of communication that isn't Quantum communication it's classical communication because I think of breaking into short term taking a big messes lots of you know lots of communication and lots of stuff flowing around inside of a cell and trying to break it into like you were saying before that that sort of Chunk size apis that communicate over a somewhat minimal amount of of you know wired signals doesn't sound like a Quantum problem to me so how does that how does all the the the mechanism the formulas of the plot the mechanics help I just totally misunderstanding well no I'm using the quantum formalism as a particularly simple way of talking about communication but the other thing it gives to you is a localization of thermodynamic requirements and so a localization of thermodynamics free energy requirements to only the places where you're encoding classical so one motivation for doing this is this analysis that Mike and I did a few years ago of to what extent can you represent the state changes inside a cell and in particular protein conservational changes as classical computation in the way that they're actually represented in say classical molecular genetic simulations where you think of proteins as sort of exploring this this configuration of State space and it turns out that cells lack the free energy resources to do that by orders of magnitude so they have to be using some other resource some you know coherence resource that's energetic and free to do some of this computation so one question motivated by that is where do they have to store Quantum information or where sorry where do they have to exchange classical information that really is energetically expensive and that they really do have to put a lot of resources into and so they're they're kind of obvious answers in the cell you know wherever proteins are bound to a membrane they're transmitting of classical State across the memory to whatever information processes that's happening on the other side that they're probably other boundaries that we can't see just by looking at say an electron micrograph with their classical apis but at least it raises that question where are the apis inside you know and how can we cut this up into modules that we know have to communicate by some sort of classical information Exchange given that can't be happy so anyway so that's just a very brief summary of Where we've been in this now multi-month course and now we're finally to um some of the part that's really biologically interesting to do with space but before I talk about space time and this is actually motivated by questions were asked after one of the other sessions and Mike seems so what is immediately to follow before because I shared some of the previous meeting it was the last few days I wanted to talk a little bit about the explanation because it raises it an interesting question about the scale dependence of biological processes so let's start by contrasting uh reductionist and scale from theories so um the sorts of approaches that we learned in undergraduate school in undergraduate physics and chemistry for example you know there's stuff going on at the molecular level and the stuff going on at the cellular level is just some emergent process from all of that chemistry with a scale free type of theory like does theory underlying this undulated by the free energy principle which is a scale free description of how systems of any type communicating with their environments and build models of their environments and behave in their environments so in any kind of reduction is theory the basic idea is the explanation all the all the real the important stuff handles happens at some preferred state and whatever's happening at larger scales is emergent or LP phenomenal or something like that it's you know it doesn't matter at all or it's some sort of Packaging of what's happening at the lower space so the fundamental assumption is that there's a fundamental skill which that's most important whereas in a scale free Theory of the approach is very different skills three Theory positive rates that there's some kind of Dynamics and that Dynamics happens at every scale right yeah and as you move between scales what you're seeing are different implementations at the same Theory so it's a very different way of thinking about Theory and I would suggest that a lot of science is actually not reductionist even though scientists talk about it as if it was rejections because I think they learned extreme but they should say and the scale-free approaches kind of more difficult to think about citizens so if we go back to that picture of systems talking to each other across the boundary and you ask about scale then the way to represent scale is the density of which information and we can think about a whole spectrum of scales that are already given to us by physical theory for example energy scales or space spatial scales and temporal scales and all of those are coupled together by invasive physics or basic physical constraints so the kind of smallest possible physically definable scale is the plank scale where and energy of about you know the 19th gdv so 219 billion electron volts is uh compressed into a spatial scale of 10 to the minus 35 meters so that's that defines the boundary of a black hole basically anything that includes information at that scale is a black hole and 10 to the 19th gev it was you know what is it about 15 orders of magnitude larger than the energy density that we can get with the Large Hadron Collider so it's a very high energy density compared to what we could produce in the laboratory so the scale about one GED is the scale at which nucleons exist and that corresponds to a distance scale we're about two minute minus 50 meters so nucleons about 10 20 degrees size um and our scale our natural scale is about meter and the associated energy is very low that's radio frequency energy so if you think about white about 10 to the fifth higher energy than radio waves so light feels energetic to us and radio waves don't and that's just a fact about our size that white seems very energetic compared to something I created which is our natural skill so we can think about the boundary as as encoding this information practically different skills so now how do we feel about science and we're all familiar with with uh kind of reduction in science where we assume the fundamental scale and this is the best example right you have some event that happens at some incredibly tiny scale in space and time and very high energy and that sets all the parameters for all the physics and produces all the organized structure that we see in our work I imagine and that's called The Big Bang Theory so the Big Bang Theory is is in a sense the ultimate reductionist theory that says that all of the important parameters that characterize everything are up to boys and nothing else fixed up the plant energy the plank Time by spatial skill and everything else just happens so what we're looking for is a way of thinking about science and scientific explanation it's different from that and it's hard to find an example that with this sort of nice Illustrated picture so we have to draw a picture and in the kind of notation we've used here um the picture looks sort of like this we've got this system and the other side of the boundary and that system is changing through some kind of God's eye type record and we're looking at our system Alice is looking at the boundary where else can see encoding is the different scales and let's call those scales one and two and what's happening on the boundary at these different scales is changing so if Alice looks at with you know magnifying glass that sees scale number one then she sees some changing pattern bits now she works with a different magnifying glass it seems scale number two she sees some changing pattern of goods and she can construct a theory about how these bits change over time and that could be for example some Markov process that captures the theory and then the interesting question is how do those two like so can you construct a saviente theory embedding Theory that says okay bit patterns at this scale have some correlation and there's some part of the correlation structure that's this sorry small scale that matches the behavior of individual bits at this very large scale so I construct some sort of if you will with it now the language that seems natural at this field and all I use this International intensity and it's how those theories change through time that's a really interesting question and you like your these semantic backings to stay fixed because then you know what you're talking about as you progress through time to make these different types of measures so this picture should look familiar because this is how computer science works yeah and so we can translate all this into to the language of computer science now the scales of different programs um which may be things like an operating system versus user interface versus what we call the hardware level which really isn't the hardware level it's a high level description but a bunch of stuff is doing we can keep going down in space with a laptop or something other scale is kind of these are interface here but what we're doing in Computer Sciences constructing exactly this kind of selectric value between different ways of describing one process and we do that by assigning meanings and mapping those meetings into each other so that's why I call it a semantic happened with computers we design them so that the technical so you know I talked about the relationship between you know the C compiler HTML or something in a way that doesn't change what we can then ask is and here's the where we start to get into biology uh what's the actual relationship between these embedding theories in biology if you think about biology in this scale-free way which if we think in a free energy principal framework we're doing oh then we can ask what the embedding theories look like as we go from thinking about how we have the pathways into cells cells in the tissues and tissues of organisms or organisms and social works and communities and what all the way up to the biosphere but we have all these different things and this gives us a set of scale transitions and if we want to represent sets of scale transitions we can turn to again mathematical formalism and representatives and renormalization which is just a set of operators that move things between different scales and we can ask how different do these transitions between different skills and so I'll just put on the table a hypothesis to think about which is the hypothesis in the biological systems those scale transitions all look like the same thing they're all formulated and I don't know whether that's true I don't even know how to approach trying to prove that it's true or not true but if it was true it would be really cool because if you think that the theory was super super simple you'd have the free energy principle and one scale operator and with that you could do all the events so I'll just leave that there's something to think about um but be that is May we've done something really important here which is we've linked physics to computer science and we've done it in a way that [Music] is not just pre-constructing service some God's eye view of physics with we're trying to construct physics from the point of view the embedded agent in a way that that looks like computer science and that immediately raises a problem which is self-reference and we know where that gets it gets you to girls there under very very general assumptions so uh girdle warns us that this theory is going to end up not being both consistent um that's that's the end of part one um now what I want to do is actually talk about space and do it from this kind of scale free perspective so ask a question yeah on that last slide your what is the semantics level It's Magic mapping on a computer it's like we have programs that are actually basically mapping that output some interpretation of the hardware other semantics and other programs like mapping what was that house looking at the program what um think about processes like variable binding right as as soon as you get up to a language where you can talk about variable marketing um if if I have a variable and I'm binding some some structure to it and I can also bind that variable to other kinds of structures then we can think about what that structure means to me I have some some notion of what two means reasonably from the point of view of the program of at least two means something different from three because if if I find a variable or two then I do something different if I'm buying the variable three and the basis of semantics you know go back to Gregory basin is if if there's if there's a difference that makes a difference to me that's invincing substance actionable That's the basis of semantics it's the basis of meaningfulness so you have some at least minimal sense of meaningfulness because as soon as you have variables a distinction between variables and constants and you have one structure that that leads to different actions a new different circumstances so the other the other way to think of it is when we are describing what programs you're doing we are naturally ascribing this and I did through the program so we're imposing a model Theory in formal terms on this bunch of syntax on this language and when we look at different kinds of languages we put very different model theories on top of them so the language is you know naturally have different instructions it was a model theories we use or typically very very different so it's that's very strict down versions referred to there is some voices in computer science that don't have distinction between software and hardware and and I'm asking that because here do you require that the hardware should be a universal tuning machine or no so it can be any emulator so I mean if we look at biology it's all Hardware right since um yeah it's just we're we're describing the hardware of different scales and we're building the hardware right as we go nuts that's even more strong statement if the how to should not be not necessarily need to be Universal yeah I mean I in a sense if you if you take this informational information Theory way of thinking about physics right seriously you get you know back to the grandfather John Wheeler and this Proclamation that physics is information Theory so Hardware is software right and that underlies a lot of that's much of the underlying philosophy of this whole transition that's occurred in quantity or in the last 30 years what's happening what's that like that John Archibald wheeler who was an associate Baltimore and Einstein so he kind of combined their different ways of thinking and he was the person you sort of I think he came up with the name black he was part of that to me Rebirth of general relativity in the late 60s and in the next six weeks go to a sequence of papers that can be seen as the founding papers of qualification motivated a lot of those people who were working early on in that area okay so let's talk about space time and really easy place to start because one of wheelers it was you have to start physics without space-time figure out where Spaceman comes from so there's the massive departure from the kind of Einstein viewed space-time is a physical fundament even though my son also said from a philosophical point of States like this that is somehow observed so this notion of space finding Observer relative runs deeply through current Quantum cosmology so let's think about space and let's think first about absolutely uniform environment nothing that's happened but so I look at the environment and none of it looks any different to me and nothing's going on so now I'm in general yeah boundaries it's not all the same yeah so so this is where I want to start um an organism that's living in an environment like this is obviously not an organism and not living um because nothing's going on in this environment but we can think of this as an abstraction okay how do I build up a meaningful environment from this and how do I start putting a space-time on how do I put some sort of coordinate system on this environment so that I can tell about you know what's going on over there so the first thing that you've got to do is somehow segment the environment in the parts that are different so let's say that we can start to segment the environment so that at least different parts of it look different even if nothing is happening so we have some part of the environment what's different from some other part of the theme park now we can ask is there any sense of of relationship between these parts and obviously the way that I've drawn it um visually to us there's a relationship but I haven't said that there's a relationship so right now actually I just have an environment that's a set of two parts and we can make it a little bit more explicit by going to Four Points so at the top I just have a set of segments and I haven't said anything about what segment is next for example even though visually I obviously wireless access you can just think of this as a here or a pile I can take this set and I can add some structure to it by adding some topology and the simplest thing to add to it is some kind of graph structure so I say okay this piece is next to this piece this piece is next to this piece Etc so I can take this and turn into a graph and I've drawn the little connections as if they were Gap Junctions to sort of suggest something about it planes of cells that can talk to each other and they know that they know they have a neighbor um there's some cells they can talk to and other cells that they can't talk to so they don't know anything about so now um there's still nothing happening even though the environment has a little bit of topological structure but once you have some topological structure once you you have more than just a set of distinctions you know the structures of distortions we could talk about something happening so let's talk about something really simple simple um there's something happens in part of the environment and then something happens in some other part of the environment and I can tell these parts of the environment are distinct because I I've broken this symmetry that allows me to move the parts of the environment around by making some things connected and other things not connected so we can represent this in a really simple way we can say in the top panel there uh a black dot has appeared in the white green Square and I represent this by an up arrow and a green subscript and a black dot and this is this is actually the notation one uses in farm field Theory so one can think of this as a what's called a raising operator which just creates a field excitation in the green part of the field and what it creates is this black dot and then we can think of a different state where black dot is disappeared from the green part of the field the white green bit and has appeared in the dark blue there so that's a representation of something and these black dots are now kind of excitations of a kind of field look included blackouts so they're not yet objects in an intuitive sense their objects in the way that say electrons are objects where if I have if an electron appears over here and an electron appears over there from out of the quantum vacuum then the question of is this the same electronics that never arises it's not even a meaningful question they have exactly the same properties of their location whether the account is the very same thing is is a meaningless question so at this level of description that's to the meaning of this question so we have a space where something's happening that you can't there's no sense of there being an object in this ratio but even in this really very simple simple system you can think about uh an observer that has enough memory to keep track of this sort of thing happening over an extended period of time so an extended internal period of time because that's with respect to some internal memory of your consequence so if you've got that kind of multi-step memory you can see State transitions like this where something happens in part of your environment and then something happens in another part of your environment something happens in the first part something happens in the second part and anytime you have that kind of periodic Behavior that's a clock in your environment and it's a talk because you have enough memory to see periodic Behavior so you have that much memory you have to have an internal counter that counts experiences and labels them in your so I can say oh I've seen that state as you can say I've seen that state before and you can watch quite a few of them and you have this kind of periodic name a clock so notice what this does it does not require you can define a clock in your if you have distinguishable environment sectors that have some topology that prevents them from being arbitrarily exchangeable so you've got these connections you've got this graph structure you need some kind of flip-flop that's implemented on that structure and you have to remember but it doesn't require objective we've implemented it just with this Quantum field theoretic notion of events happening in one place that are happening in another place and you don't have to have a metric space you don't have to have any sense of distance so you've got a really really minimal structure on yours facials representation of your environment but you now have a nice external time reference uh that you can look at other events with respect to so this tells us something really important I think for biology which is that time is more fundamental respects we can expect organisms um even single cells to be able to measure time in their environments we can expect to see the measuring space within their environments and you know organisms that we know about have things like diagonal Cycles even bacteria the we don't have any real reason to believe impose a geometric structure on their even though even bacteria distinguish sort of part of their number because the whole way that bacteria you know Implement chemotaxis that's just what you said they're they're so they want to look towards gradient but they're so small that one side of their body and the other side of their body are at the same concentration for all practical purposes they cannot measure a gradient in space they measure it in time because that's what works yeah and even before bacteria I mean movements can can later write first organisms to move much that makes all the sense and at least I don't right now and truly can guess how you would measure space that movement it's going to be probably possible for some like light diffraction patterns but without that um you know they can't move it's gonna be way easier to measure because there's a time of the environment to somehow doing space it makes sense that we're involved yeah I would expect even that period you can move or even your Roblox that can move may not be representing motion they may not be representing objects stuff that's more easy to represent thesis so so now let's think about representing motion I'm sorry can I can I ask a physics related question sure how does this relate to the esotropy of space because if you compartmentalize space into different cells it does have need to have a certain kind of uh well Arrangement so that light in One Direction and in the diagonal direction for instance takes the same time to arrive at a certain distance right uh yeah let's get to that in a little bit I can talk about metrics okay okay I think that I think that's really a metric question or almost a metric okay let's think about what do you need to observe motion um what do you need to be able to say that there's some object that's moving around suppose the I had this black dot and it's it's exchanging its position between light green and dark blue but now I want to see that as motion so see that as motion I need to add an assumption which is this this black dot is not just a field excitation of some Quantum field that's actually a thing and what it means to be a thing is that it persists through time it has some identity Through Time so that I can say that when it's in the dark blue square it's actually the same thing that used to be in the white green Square and that that same thing is going to move now again at the end of the white so object persistence this this idea that I have a thing is deeply deeply related to the notion of translational invariance which is one of the fundamental symmetries that allowed you to define the space the space time of the sword that we have so we have to make this extra assumption that moving something doesn't change its identity and similarly moving something in this space doesn't change the structure of the space so I have to make this assumption this is part of translational experience that if I move an object from here to there I don't change the topology of the where we don't have a geometry yet so we have to assume that we still have to assume that it doesn't can change the connectivity structure in my space to move something movable so what I want to emphasize here is that this notion of object persistence is a classical idea well it's it's not the idea that arises naturally in Quantum filter which says simply that something appears something else appears that they have no persistence I think we can think about the entire history of 80 years of interpretations of quantums but you know hundreds of books and thousands of Articles written discussing metaphysics basically has all been really a debate about object persistence and what the meaning of object persistence because as soon as you have problems with persistence you made the transition from a purely quantum theory to a theory that's quantities object persistence is what really defines classical information it's I can write a bit I can write here and

I can come back tomorrow if I can read it and there might be some noise that often the noise it's the same bit that I'm going to do it for That's a classic car dude so here's the quantity classical transition it's the Assumption of object I think it's worth thinking about where that assumption comes from Chris can I ask a question if you can hear me um so if you go back to the previous slide uh could we also not in a sense think of these sectors the green blue sectors as also as things because they persist even though they do not show any motion could we not think of those sectors also as things because of resistance I I represented them this way to to have a visual depiction um but the the intuition I'm trying to get across is an intuition of of location or distinction between different parts of the environment so we don't tend to think of you know this little piece of space is an object you know a little some little voxel somewhere between me and the window we just tend to think of that as as part of our coordinates so I've I've drawn it in this visual way which suggests object to it but you know the very uh kind of a voxel in my space is really different from something that I could put into that box so so this thing that I can move around as an object whereas the kind of voxel is an abstraction so yeah I don't think they're the same thing at all I I guess uh I'm not an expert in physics or relativity or but doesn't Einstein's theory of space-time say something about how space-time could be deformed or marked or stretched and contracted so doesn't it suggest that space is also an object in effect we'll get to that in a couple of slides okay okay yeah hold hold your breath on that one so so I want to I want to talk a little bit about object persistence more and and just and and let us think about that from kind of a developmental psychology point of view this is how we learn about object persistence and we we get this about the first three months of infancy we do this kind of motor babbling and we do it with objects and we don't have a a handy colorful thing we just look at our own limbs and notice what's happening here in this picture of illustrates doing well there's visual attention and there's also motor attention so what we're doing when we're doing this motor babbling is we're coupling a somatosensially inappropriate safety of representation of our body with a visual representation of our body and the visual representation is assigning object Hood to this thing that's otherwise a motor representation so what's a motor representation it's a among other things is the amount of energy I have to spend it's effortful to do this and I can sense that effort so and this is this is very indirectly getting a percentage of this question I'm starting to couple the ideas of energy and object by doing this sort of thing and I'm also coupling it to the idea of persistence over time and manipulability so these are deep deep intuitions that we have and uh I think we can say just based on physiological analogy as well as behavioral analogy these are deep deep intuitions that animals have and maybe even the vertebrates have so we can think about pretty easily from the point of view of second dog uh and maybe even we can think about space in the point of view of some frog or a fish um but we don't have a good way of at least I don't have a good way of thinking intuitively pronunciation of point of view something like an insect it has a super different developmental process than more and isn't obvious in doing this kind of process that was the process that taught me what space was uh I think we we need to think about that when we're thinking about space in terms of biological systems so let's just in the final segment here think about what it takes to start to impose a geometry on specs so if I want to build a coordinate system I want to build a geometry in our area of motion I need to have motion with respect to something so I need some sort of designated preferred position but I can say okay I'm moving relative to this and as soon as I've got that I can start to build some other invariances so I can start to think about for example rotational invariance because now I can tell I'm rotating because I can say okay this is approaching but this is rotating so now I can think about rotational variance in the same way the thing about translational variance and if I want to think about three-dimensional space I can think about kind of size so if I if I move something it gets smaller and if I put it closer to here it gets bigger I learned that when I was three months old because I had this mother getting ready for me um you can also think of that as as a kind of energetic intervention so if my object is over here I have to spend energy to get to in a way that I don't have to spend that energy if it's right here so you can think of this building this third dimension in various ways but you always have to have this origin to build any kind of geometry so once I have this I can also do something really radical which is Imagine like 20. and uh I think it's an interesting question to ask what animals are capable of representing themselves as opposed to representing their environments is moving and I don't know how to answer your question I think it's a really interesting question so uh finally I don't just need an origin if I want a coordinate system I have to have a ruler so I have to have a spatial reference for that consists of this origin and some will work and my ruler has to be translationally and rotationally and very up to some transformation and and here we get to the relationship between say euclidean experience where the ruler is invariant to translations and rotations um and and something like space in general relativity where if I move my ruler around it changes length and if I rotate my rule of a chain easily and that length change is a representation of mass in general Santosh was pointing out so in fact it's it's not that this space is a representation Beyond change it's the change in the metric that represents mass in general relativity so it's the way my ruler changes as I move it and that's a I think that's a very interesting concept which we'll talk about so once I have these things once I have the spatial reference frame now I can talk about objects moving into geometry and I have what what we consider intuitively in space so notice how much more complicated this is than a clock right simple to get a clock that's really complicated to get this space is a much more complicated notion than than external time so we're almost done I just want to think a little bit abstractly now interdisciplinarily now and think about the approaches to emergency I mean in the in the physics literature especially Quantum information and to a somewhat lesser extent under the rubric of quantum cosmology there's a large industry for hundreds thousands of papers uh they're working on emergent SpaceTime and it's it's all an in a deeply technical difficult language and this route involves many many variants on starting from some sort of information exchange and usually entanglement and working through the idea of error correction uh in error correcting codes and through the idea of lattice-like so discrete Network types of topology to get to some sort of space-time and in particular by always adding a bunch of different kinds of assumptions to get to Einstein's equations to to get to a space time that supports some idea of mass of gravity or energy density I mean all of those are equivalent so some sense of a metric that changes when you do certain things now there's a completely different approach which isn't expressed in incredibly complicated mathematics let's Express in the language you developmental psychology that way of thinking about space-time starts with perceptual motor capabilities so it starts from from motor family uh and it talks about the measurement of effort so it assigns an energy coordinate at the top and it talks about memory and it talks about objects uh and invariances so transcribe the ability to move something around it and say this is the same thing that I'm going to change blah blah and from there it builds this notion of a metric space and that metric space has a recent implicit energetic coordinate that has to do with distance um it also has sort of an energetic component that has to do with mass so I know that there are things in my space that require a lot more effort to manipulate than other things but in in this way of thinking about it that notion of effort in mass is not really anchored in the space it becomes a different Concept in terms of this classical notion of mass and the question is do these routes eventually produce the same thing the the sort of obvious answer is no right the obvious answer is that uh the left hand side this this very recent you know two decade two or three decade definitely theoretical physics that starts moving from string theory how produces this this very abstract notion of space that gives you Einsteins equations but is that really true I mean one gets general relativity just by thinking about symmetries in our ordinary conception of space that people haven't really focused on for right Einstein really derived general relativity from the equivalence principle which is the idea that acceleration and mass are the same did it with all these thought experiments of people in elevators and things like that so in a sense general relativity follows from our classical thinking about space we noticed some symmetries that you haven't noticed before and you wrap some mathematics so maybe these paths really do converge and I think if they do converge it's really interesting um because it tells us something about the relationship between physics and that I think is important um what I I hope I convention today is that there's at least some possibility that maybe these two paths really are leading to the sense that really we can think about constructing space time from an organismal point of view so I'll just leave you with one picture which is a organisms and babies and systems that aren't organisms managed to construct this cycle from memory to object persistence to error correction which I really haven't talked about much again participation um I'll just make one comment that that space is really useful because it means that I can I can correct errors again put things in space and expect them to be there and that gives me an external memory so I have space I have this converging memory I have a whole lot of American functions I can use my ID card remember my name oh so so organisms can do this and that raises the question which I think we can view as a phylogenetic question of how did this start um where in phylogeny does this cycle get going and how does this cycle change in different parts of phylogeny how do different parts of phylogeny implement the site and I think that's a really interesting question in evolutionary biology that's what we're thinking about so I'll close with that thank you very much I'd like to build anyone [Laughter] so we're we're basically at one o'clock so if there are a few questions that's great you my next step is to go to the airport soon you remember the experiment that's not allowed to do today is there is another experiment in that category there were some counselves that took Keaton connected with a hinge together Moon one of them it's a concept that it's completely closed and there is tribes in that car so and one kitten is able to use these nerves and to move but his movement affect the other kitten that they folded and they're not allowed him to use the legs and they they put that kitten from from the beginning like the result was that that kitten was blind and there is nothing wrong with that kitten only because he never used his legs to close the loop with the environment okay so his movement were only completely not connected to him it's the other cut so they released that cut later on I don't remember exactly the time numbers and the how many months they put it there but they releasing and they let him use the the the the legs and he gets back his eyesight so maybe that related to this kind of thing yeah I bet no one can do that experiment today but yeah yeah working one can easily think of experiments would be incredibly informative and completely certain importance I like this because I I feel like yeah this at the molecular level right and and I think you can think of space as being any area with less resistance right so if you were to Define what space is in this room most people would say well it's the air right this is the object whatever it has more resistance but I think a molecule does this too right because as molecule binds to another atom and it changes it has a memory of what it bound to and now you have object persistence right because now you have a different structure there's changes the way the thing interacts all the way through and it has error correction because it doesn't bind it it won't bind to a molecule that doesn't have the right you know valence electrons around it so you know once that bumps into each other it will reject it out and then it'll grab the other one right so I think you're seeing all this you know not even organisms next I mean yeah I would love to be able to do psychology and really think about these things as agents that are in some sense common yeah absolutely yeah we're all crystalline lattices that's all we are you know all the information that we're transferring and thinking about all that is being transferred through Crystal and lattice yeah so what will be the memory in that kind of thing the structure now the structural memory and I think there's there's in a sense false has converging them and the molecules can write on their environments in ways that are persistent and in ways they that they can go back to and get information at least I would have problems obvious that molecules are doing that certainly organisms are doing that all the time yeah it's a cool experiment in field mice when they leave like literal markers on the ground that are visually distinct from the rest to remember where to go I mean the word came originally from social insects or memory of the environment if we you know molecules using stigmerging memories would be a super cool result but you can really nail it in here makes me wonder if uh is like a basic kind of foundation for cognition you know when you mentioned like you know basically all of biology it is chemistry it's just physics interactions based in most surveillance of electrons relationship between those two but that's basically all the molecules too and to extend positive and negative charges both positive and negative experiences so so Mike's here is taking notes to talk to Mark songs about this yeah I think there's also good examples of all the stuff that you could put together with like systems Neuroscience as well which speaks to your first point of the scale free stuff right it's probably all the same like I I'd be more surprised if it was people that I give up yeah I think it's true you came here to give this course um well I came here to see who I haven't beat people are you standing around for a bit or do you have to leave like right away from the airport okay can I ask a last question maybe an iPhone why is there three special dimensions I'm sorry can I ask a naive question um why is there three spatial dimensions and not two or one for instance is there a particular reason to that got those they're they're similar medical facts about three space that I've never been around that are real interesting yeah like building knots or something like this I mean you cannot do everything in two Dimensions or also in one year but um but there are also things that you can do in three dimensions that you can't for example do in four dimensions ah like so a you know four dimensions plus time would be uh would not allow certain symmetries that we depend on too um to kind of have some coherent notion of objects so and you know I'm sorry I don't remember what these things are but uh that's a really interesting question and a lot of people work on it and um an important part of it is is why three not four or more and and three turns out to have unique unique geometric properties that are really cool so it when people talk about extra dimensions and strength area or something like that all the extras facial Dimensions after we rolled up into tiny little packages so that they don't interfere with with doing anything macroscopically where macroscoplings you know above the plank scale yeah yeah this is really interesting um also if it's about um well distinguishing between objects then also there's questions if space can be continuous for instance because making very very small changes in a continuous we won't lead you to different positions right so yeah this is where the the limitations imposed by energy density really make a difference you know the playing scales tiny um if you if you try to go to go beyond the the energy densities are so high that it it sort of feels like you're getting big again um it's again I don't have a good intuitive grasp on that but the the existence of this energy distance kind of gives you a fundamental dissertation of space for free and uh when you look at these Quantum information types they're they're all based in five-dimensional over spaces and there's been a huge transition away from the kind of old quantum mechanics where everything was done in continuous over spaces to the much much simpler representation of doing things and try out dimensions uh the downside of that it means that that all of continuous mathematics like differential equations becomes an approximation on the Dynamics that's conceptually simpler but but from a computational point of view much more difficult yeah it's it's really hard to grasp um I I when I studied this stuff I always imagine space time to be some kind of uh automaton where you have like a little space grids and uh well communicating with each other with the speed of light something like this that being the clock time of the of the automaton but it feels like it's much more Dynamic and Space is really emergent and more like a property you assigned to particles or anything that distinguishes them from each other and it allows you to build correlations on top of other degrees of freedoms